

IS492 Spring 2026
CheckPoint 2

Injury Aware Workout Adapter

A workout adapter and generator that helps users modify training plans and avoid secondary injuries

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GitHub repo link:

<https://github.com/IS492-SP26/team-project-injury-aware-workout-planner>

Validation

	CP1 Assumption	CP2 Reflection
Emma	Basic injury information is enough for workout adaptation.	Injury-specific guidance and pain management instructions are needed.
Prisha	Are LLM generated alternatives grounded any manner	AI tools have been fed massive datasets which even doctors have come to start trusting
Vinit	Self-reported injury information is sufficient.	Movement-based assessments are needed for reliable evaluation.
Ocean	Direct workout adjustments	Risk warnings and reasonings behind those adjustments are needed.



Feature List



Detailed injury survey

Pre-survey on user injury status.



Training plan adjustment

Plan adjustment with risk level warning and adjustment reasoning



Various plan formats supported

Videos, texts

Prompting Study Design

✦ Prompt Cases Validation

Same input text for different LLMs.
Same prompt structure.
Compare output quality.
From constraint → no constraint → video test

✦ Existing Tool – InjuryMap

Provides guided injury assessment.
Used it to create a demo user and
understand user journey for an
injured person.

Tool → Strength → Gap → Design Opportunity

✦ Tool Validation for Market Needs / Reliability

Current fitness apps either generate generic workouts or provide fixed rehabilitation routines, but they do not adapt real-world workouts to a user's injury condition. Our tool bridges this gap by analyzing existing workouts and intelligently modifying exercises to provide safer alternatives based on injury context and recovery stage.

Evidence

1 Typical Case

Prompt: With injury constraints

I injured my knee two days ago while playing soccer. Pain is 7/10 when walking and my knee is swollen.
I still want to do a leg workout today.
Can you generate a workout?

I'd strongly advise against a leg workout today, and I want to be straightforward about why.

A 7/10 pain level with visible swelling two days after an acute injury is a sign that your body is in an active inflammatory response. This is meaningfully different from the mild, chronic irritation in your previous workout request. Training through this could:

- **Worsen the injury** — swelling indicates tissue damage (possibly ligament, meniscus, or cartilage) that hasn't been assessed yet
- **Delay healing** — loading an acutely inflamed joint disrupts the early repair process
- **Mask a serious structural injury** — sprains, meniscus tears, and ligament injuries like ACL/PCL damage all present this way and require diagnosis before any exercise

What I'd recommend instead:

The most important next step is seeing a doctor or physiotherapist for an assessment — especially given the swelling. They can rule out anything that needs imaging or specific

Output as expected, with safe disclaimer and no workout plans

2 Edge Case

Prompt: With no constraints

I injured my knee while playing soccer. I want to do a leg workout today. Can you generate a workout?

The "Knee-Friendly" Leg Day

The Rules:

1. **No Pain:** If you feel sharp pain, stop that exercise immediately.
2. **No Deep Flexion:** Avoid deep squats or lunges where your knee goes far past your toes.
3. **Control:** Move slowly. Don't use momentum.

The Workout

1. Goblet Squat (Partial Range of Motion)

- **Why:** Holding a weight in front helps you stay upright, taking pressure off the knees. Only go as low as you can without pain.

Output slightly risky, with pain notifications but full workout plans

3 Failure Case

Prompt: Video link in prompt

I want to follow this workout video today "https://www.youtube.com/watch?v=XOivdhy7BDI&list=PL2ov72VWpiOpzZHrFcq2-k8U6l-IGLa9Y&index=1&t=261s", I have a shoulder pain, help me adapt the workout plan.

Safe vs Avoided Sections (Shoulder-Strain Modified)

0:00 – 2:30

Warm-up movements (marching / light cardio)

✓ Safe

Why:

These movements mainly involve the legs and light arm swings, which increase blood flow without loading the shoulder joint.

Output give timestamps don't align with video, or even cannot access.

Evidence

4 Injury Map

Injury Assessment & Rehab Platform

What Worked

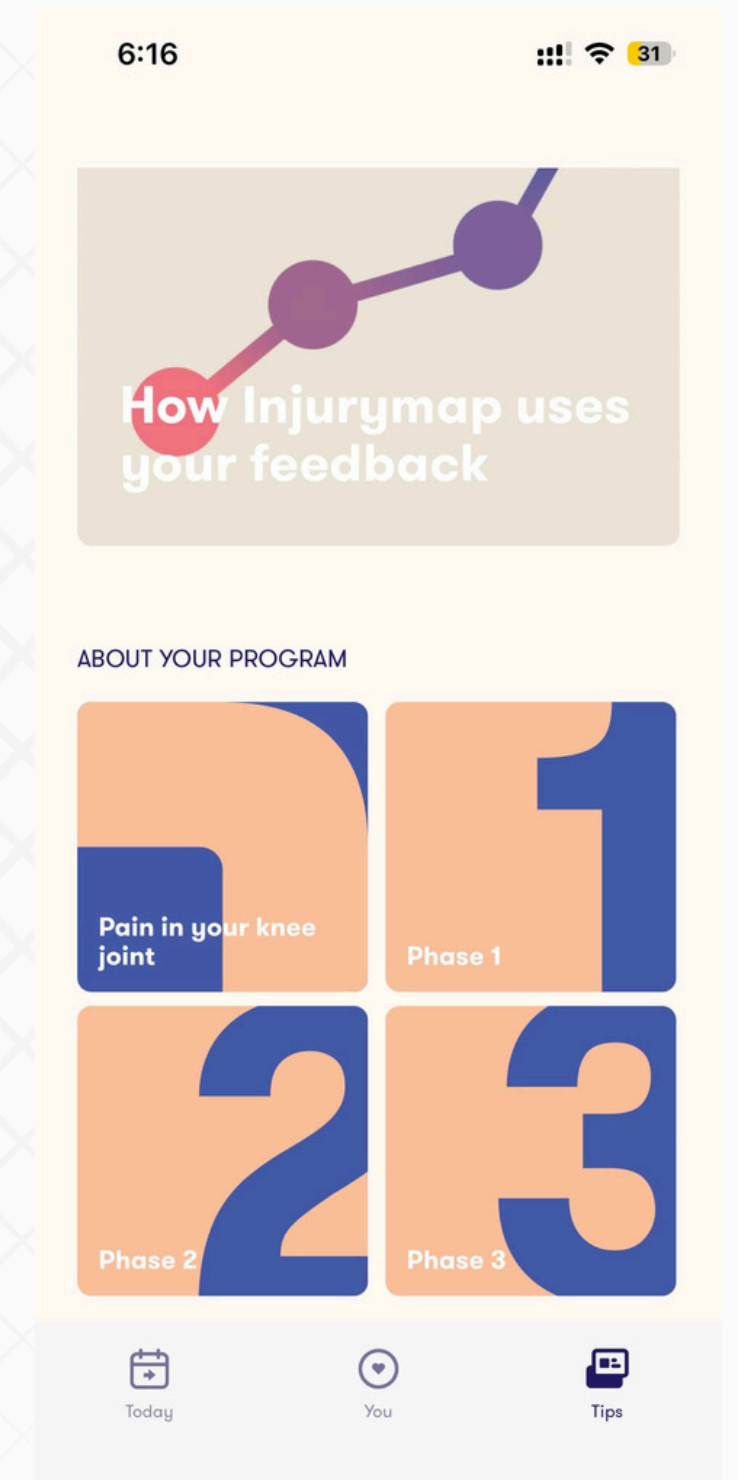
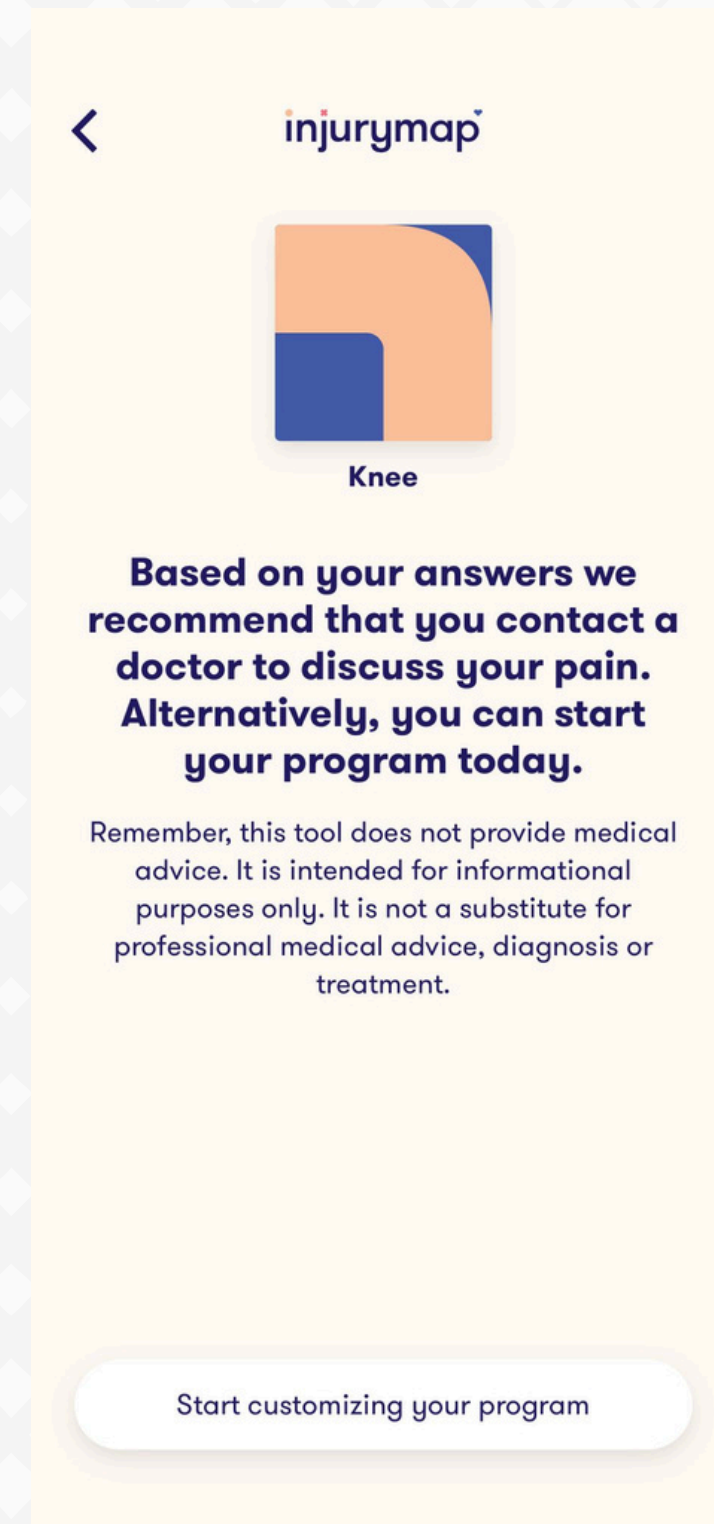
- ✓ Provides initial injury assessment before exercises
- ✓ Warns users to seek medical help if injury appears severe
- ✓ Structured rehabilitation phases based on injury type

What Failed

- ✗ No dynamic “generate today’s workout” functionality
- ✗ Programs are fixed phases rather than adaptive to daily condition
- ✗ Workout details locked behind paywall, limiting usability and transparency

UX Friction

- Requires navigating multiple steps before seeing exercises
- Limited ability to customize workouts based on goals or training style



Gap Analysis

✦ Existing Tool Limitations

No Adaptive Daily Workout Support

Current apps in the market either don't support injury assessment or give long duration rehab plans that are injury specific.

Limited Format Support

Limited ability in analyzing the videos by timestamps.

Need Detailed Injury Info

If prompt has no injury constraints, LLMs may give risky workout plans.

✦ Requirements

» **Workout Adaption**

Adaptive daily workouts based on injury assessment and recovery stage.

» **Video Analysis**

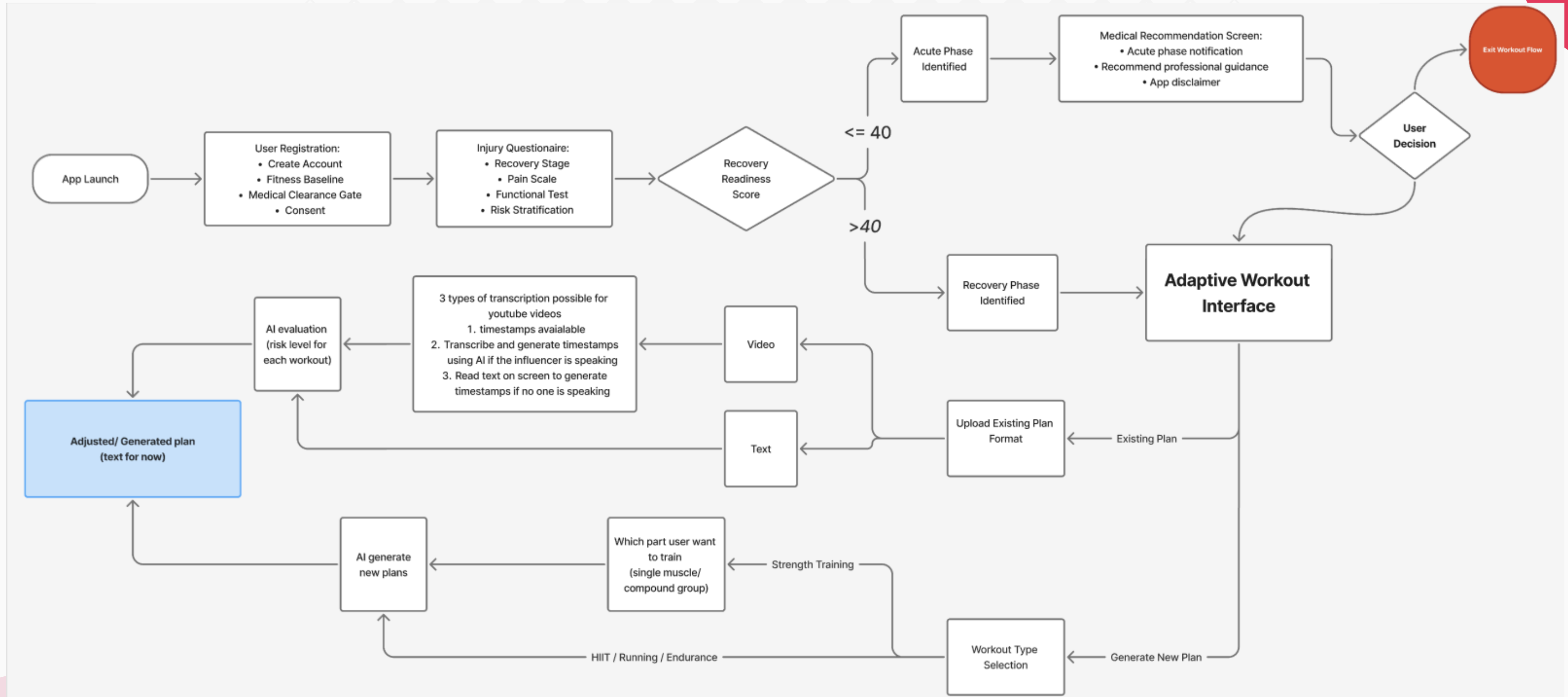
Professionals aiming to expand broader career opportunities globally.

» **Injury Survey**

In user registration, a survey includes injury level and duration is needed.

Our Task Flow:

[Click to view UI prototype](#)





Risks, Safeguards & Path to CP3

Risks

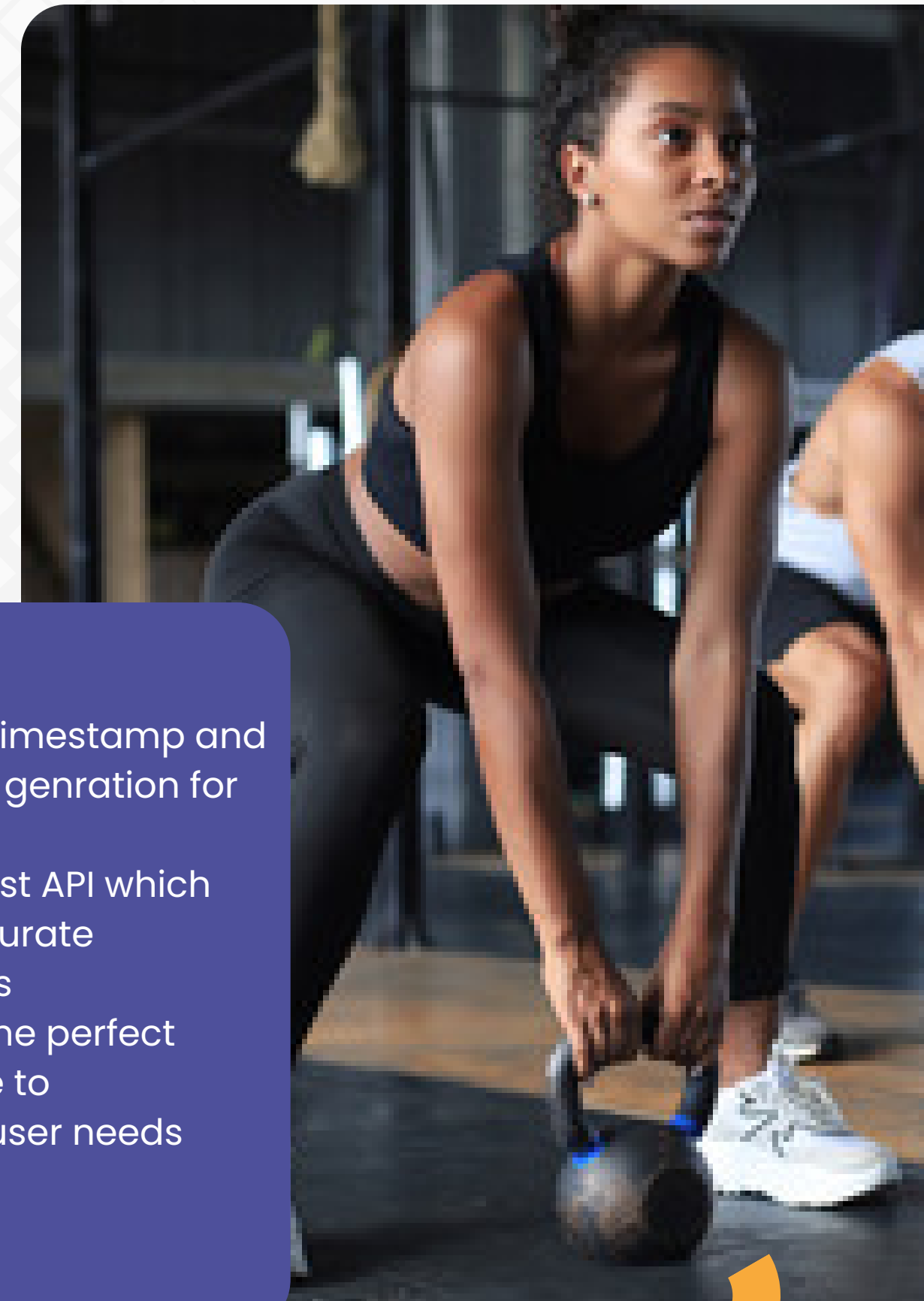
- AI recommendations may be unreliable for injury-specific exercises and could misinterpret workouts from videos.
- **Privacy concerns** when processing YouTube content or user injury information.
- Transcription and video analysis can introduce **computational cost**

Safeguards

- Add medical disclaimers and recovery-phase checks before generating workouts.
- Ensure user data is minimal and anonymized, and restrict processing to publicly available video content.

Path to CP3

1. Understand timestamp and transcription generation for videos
2. Select the best API which can give accurate modifications
3. Generating the perfect questionnaire to understand user needs



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Thank You •

